

Subscriber Retention Strategy: A Data-Driven Churn PoC for Company A

GCI World — Final Assignment

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How I understood the assignment

- I read the brief as: Company A has rich telecom data but needs help turning it into action.
- My job is not only to train a model, but to connect patterns in the data to a clear business problem (churn) and a practical next step (who to contact first).
- I kept methods straightforward on purpose so results are easy to audit and reproduce in my notebook.

Business problem in my own words

- Churn means we lose recurring revenue and pay more to replace customers later.
- If we can estimate who is more likely to leave, care teams can target offers or handset upgrades.
- For this PoC I treat churn as a yes/no prediction task using past behaviour and account fields.

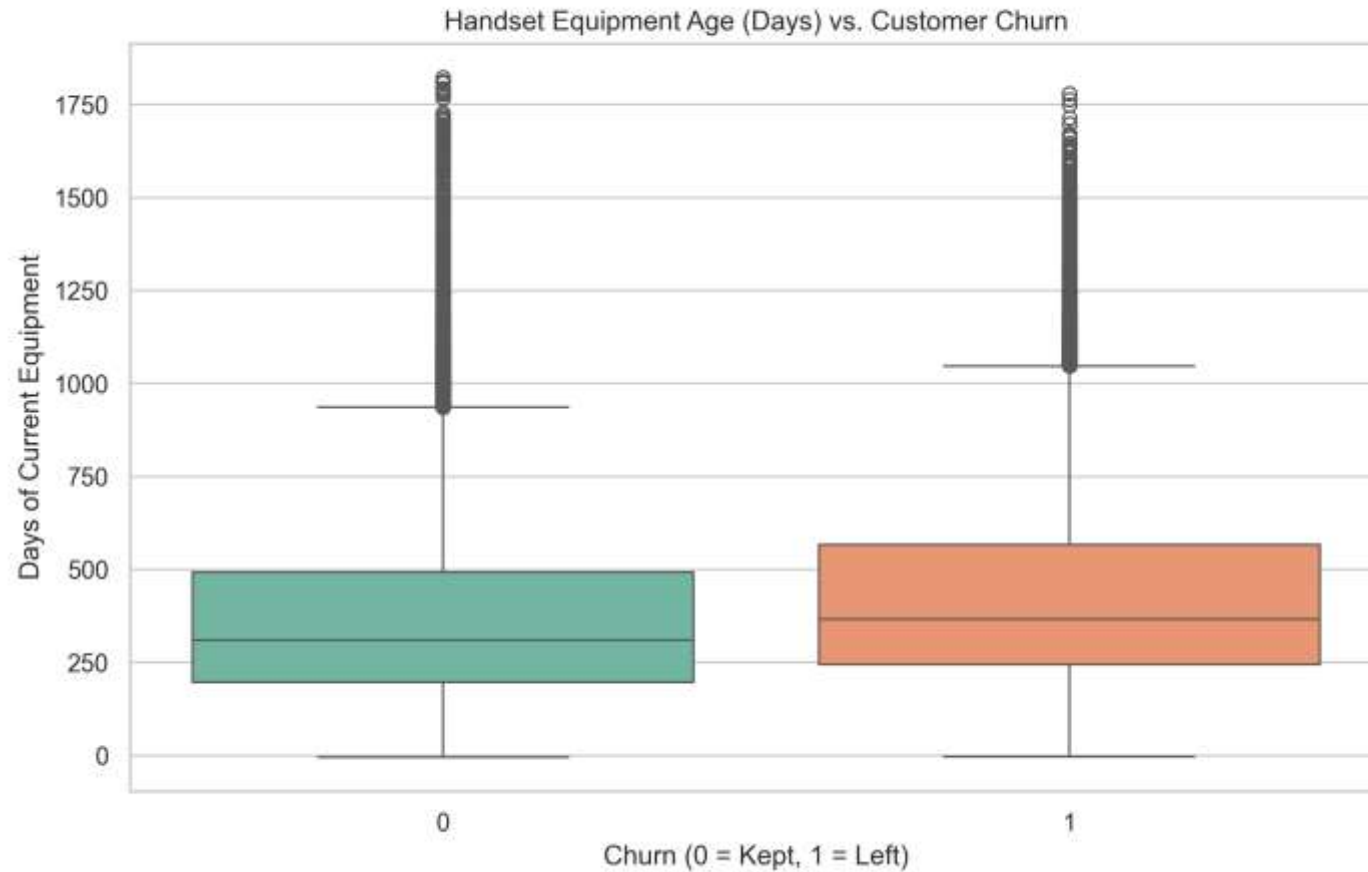
Data I used

- Client table: household, handset, revenue summaries.
- Record table: monthly usage, drops, change vs last 3 months, and the churn label.
- I merged on Customer_ID → 100,000 rows. Rough churn rate: 49.6%.

What stood out in EDA (before modeling)

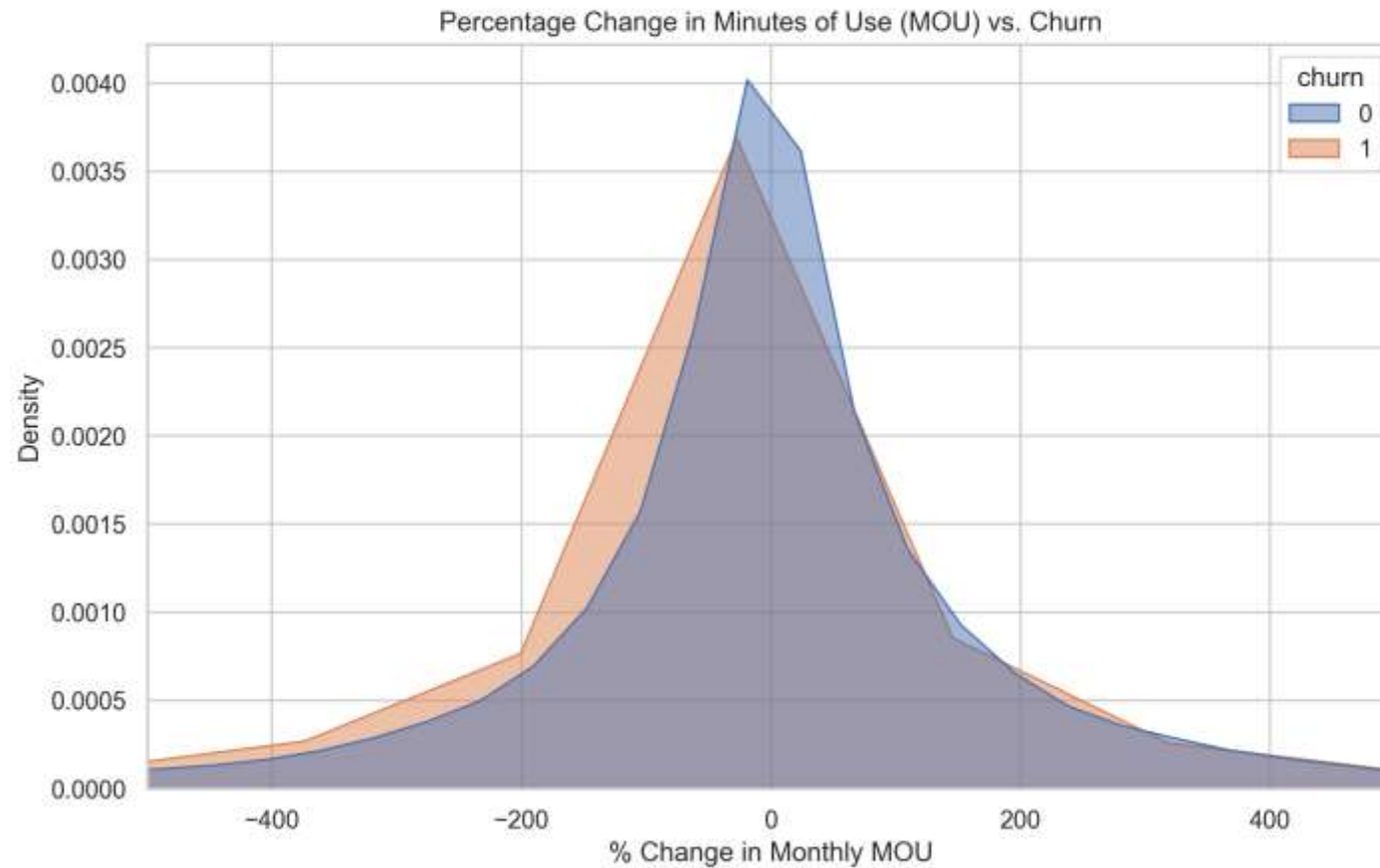
- Customers who churn often show older current equipment (more days on the same handset).
- Sharp drops in minutes of use also appear more often in the churn group.
- The next two slides are the same figures I saved in my analysis notebook.

Chart 1 — handset age vs churn



Boxplot: churners tend to have older equipment on file.

Chart 2 — usage change vs churn



Density plot: churn group more often has large negative change in MOU.

Models I chose and how I validated them

- Models: (1) Logistic Regression — simple baseline, easy to explain. (2) Random Forest — handles non-linear patterns without much manual feature work.
- Validation: 70% train / 30% test, stratified split so churn rate matches in both parts.
- I never tune on the test set; it is only used once at the end for an honest score.
- Main metric: ROC-AUC on the test set (works better than raw accuracy when churn is uneven).
- I also report recall on the churn class at the default 0.5 threshold to show catch rate.
- Preprocessing: median imputer + scaling for numbers; most-frequent + one-hot for categories.

Models and scores (test set)

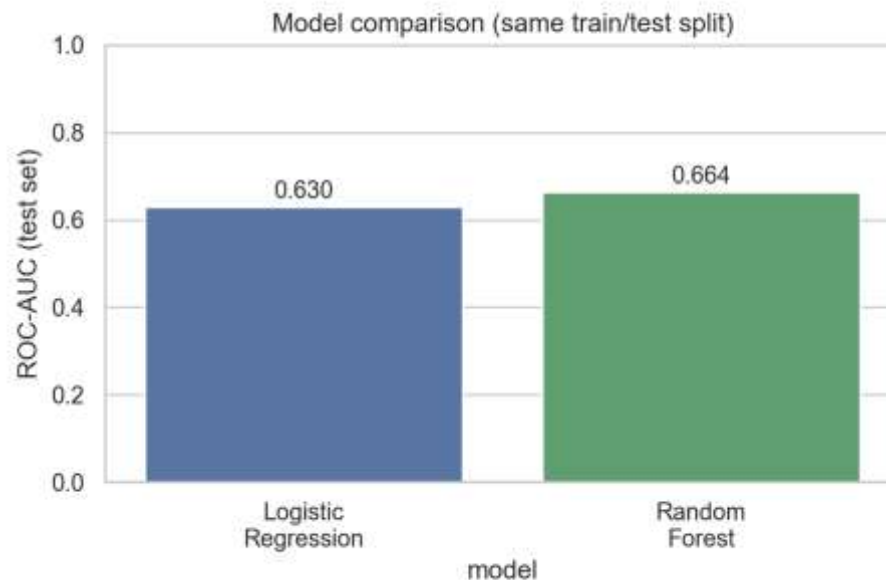
Model 1 — Logistic Regression

- Task: predict churn (1) vs stay (0) • Main metric: ROC-AUC
- Score: 0.6296 • Recall (churn): 0.5954

Model 2 — Random Forest

- Same task • Main metric: ROC-AUC
- Score: 0.6637 • Recall (churn): 0.6471

ROC-AUC: higher = better ranking of who is likely to churn.



PoC recommendation — which ML I would use first

- I compared two sklearn models on the same test set. The higher ROC-AUC was Random Forest (0.6637).
- The other model (Logistic Regression) scored 0.6296 on ROC-AUC.
- What I recommend for this PoC:
 - Primary model for the risk-ranked outreach list: Random Forest.
 - Reason: ROC-AUC measures how well the model ranks who churns vs stays; for retention calling, ranking matters more than a single accuracy number.
 - Churn recall on default 0.5 threshold (test): Random Forest at 0.647 vs Logistic Regression at 0.595 (I still report ROC-AUC as the main metric because churn is imbalanced).
- Backup / communication tip:
 - Logistic Regression is easier to explain line-by-line to non-technical staff; I would keep it as a secondary check, but not as the main ranker if it loses on ROC-AUC.

Simple ROI sketch (illustration only)

- Assumptions shown in class-style examples: $CLV \approx 4,000$, offer cost ≈ 800 , 40% accept.
- Test cohort size: 30,000 customers.
- Target top $\sim 10\%$ by risk: 3,000 people; true churners in that slice: 2215.
- If offers work as assumed, roughly 886 saves $\rightarrow$ protected revenue on the order of 3,544,000 (currency as in your course example).
- Campaign cost $\approx 2,400,000$; net illustration $\approx 1,144,000$.
- Numbers depend on assumptions — tune with finance before any real campaign.

Suggested next steps

- Run a small controlled trial (offer vs control) before full rollout.
- Track contact rate, accept rate, and complaints.
- Refresh the model quarterly as products and pricing change.

References

- Course dataset: GCI World — Company A telecom CSVs (assignment pack).
- Tools: Python, pandas, scikit-learn, matplotlib/seaborn.

How to run the Python project

- Need: Python 3.9+ from <https://www.python.org/downloads/> — on Windows, enable “Add Python to PATH”.
- Open a terminal in this project folder (same place as requirements.txt, dataAnalysis.py, and the telecom/ folder).
- Install libraries: `python -m pip install -r requirements.txt`
- If pip shows SSL/certificate errors: `python -m pip install --trusted-host pypi.org --trusted-host files.pythonhosted.org -r requirements.txt`
- Run the analysis (charts + model scores in the terminal): `python dataAnalysis.py`